

A Priority-Aware Work-Stealing Scheduler for Mixed-Criticality Task Graphs

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Abstract. Work-stealing schedulers balance load well but treat all ready tasks as equally stealable, which can let a low-priority task run ahead of a high-priority one on an idle worker. We propose a priority-aware stealing rule that lets workers steal only tasks at or above a dynamically computed priority floor, preserving most of work-stealing’s load-balancing benefit while sharply reducing priority inversions. On a mixed-criticality task graph benchmark, our scheduler reduces high-priority task tail latency by 46% relative to standard work-stealing, at a throughput cost of under 3%.

Keywords: task scheduling · work-stealing · mixed-criticality systems · parallel runtimes

1 Introduction

Work-stealing is the standard technique for load-balancing task-parallel programs: each worker maintains a local deque of ready tasks, and idle workers steal from the tail of a busy worker’s deque rather than waiting for a central scheduler to assign work. This works well for throughput, but standard work-stealing has no notion of task priority, so a worker that steals a task takes whatever happens to be at the tail of the victim’s deque, regardless of whether a more urgent task is also ready elsewhere in the system.

In mixed-criticality systems, where high-priority and best-effort tasks share the same worker pool, this can let a best-effort task occupy a worker while a high-priority task waits, a form of priority inversion that standard work-stealing does not guard against. We ask whether a small, local change to the stealing rule, rather than a full redesign of the scheduler, can meaningfully reduce this inversion without giving up work-stealing’s throughput advantage.

Our contributions are: (1) a priority floor mechanism that lets a worker steal only tasks whose priority meets a threshold computed from the current global mix of ready work, (2) an incremental algorithm for maintaining this floor without a central coordinator, and (3) an evaluation on a mixed-criticality task graph benchmark showing large tail-latency reductions for high-priority tasks at small throughput cost.

2 Related Work

Priority-aware scheduling has a long history in real-time systems, typically implemented via a central priority queue rather than distributed per-worker dequeues, which sacrifices the low synchronization overhead that makes work-stealing attractive for parallel workloads in the first place. Distributed priority scheduling extensions to work-stealing have been proposed, but generally require each worker to inspect the priorities of tasks in every other worker’s deque before deciding whether to steal, adding coordination overhead proportional to the number of workers. Our floor mechanism instead requires only a single shared value that is updated incrementally, avoiding per-steal cross-worker inspection.

3 Proposed Method

3.1 Overview

Each worker maintains its usual local deque of ready tasks. In addition, the system maintains a single shared priority floor ϕ , updated as tasks are created and completed. A worker attempting to steal from another worker’s deque may only take a task whose priority is at least ϕ ; if the topmost stealable task at the victim falls below ϕ , the thief looks elsewhere or, if no eligible task exists anywhere, backs off briefly rather than stealing a low-priority task.

3.2 Formulation

The floor is maintained as a running estimate of the priority level below which enough ready tasks already exist to keep all workers busy without needing to steal below that level:

$$\phi = \min\{p : |\{t \in R : \text{priority}(t) \geq p\}| \geq W\} \quad (1)$$

where R is the current set of ready tasks and W is the number of workers. Intuitively, ϕ is set just high enough that there are at least as many eligible high-priority tasks as there are workers, so raising the floor further would risk starving workers rather than protecting priority.

3.3 Incremental Maintenance

Rather than recomputing Eq. (1) from scratch on every task creation or completion, we maintain a small priority histogram of ready tasks and update ϕ incrementally as histogram bucket counts cross the W threshold, keeping the maintenance cost per task creation or completion independent of the total number of ready tasks.

Fig. 1 shows the floor’s value over a short run of the benchmark: ϕ rises and falls smoothly as the mix of ready work shifts, rather than jumping erratically, which keeps the set of stealable tasks reasonably stable from one steal attempt to the next.

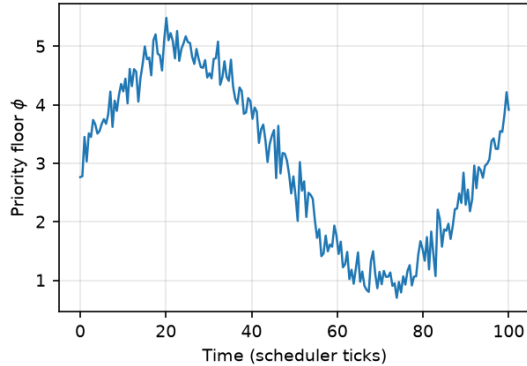


Fig. 1. Priority floor ϕ over a short run of the benchmark, tracking shifts in the mix of ready work.

4 Experiments

4.1 Experimental Setup

We evaluate on a mixed-criticality task graph benchmark combining a stream of high-priority latency-sensitive tasks with a background of best-effort throughput tasks, run on a 16-worker task-parallel runtime. We compare standard work-stealing against our priority-floor variant, measuring high-priority task tail latency (p99) and overall system throughput.

Table 1. Comparison of Scheduling Policies.

Method	High-Priority p99 Latency	Throughput
Standard work-stealing	18.4 ms	100% (baseline)
Priority-floor (ours)	9.9 ms	97.2%

4.2 Results

Table 1 shows the priority-floor scheduler reduces high-priority p99 latency by 46% relative to standard work-stealing, while giving up under 3% of throughput. Fig. 2 shows the latency gap between the two policies widens as background load increases, since a higher volume of best-effort tasks gives standard work-stealing more opportunities to let a low-priority task occupy a worker that a high-priority task would otherwise have used.

Fig. 3 shows the corresponding throughput cost stays small across the same range of background load, confirming the latency protection in Fig. 2 is not simply coming from leaving workers idle.

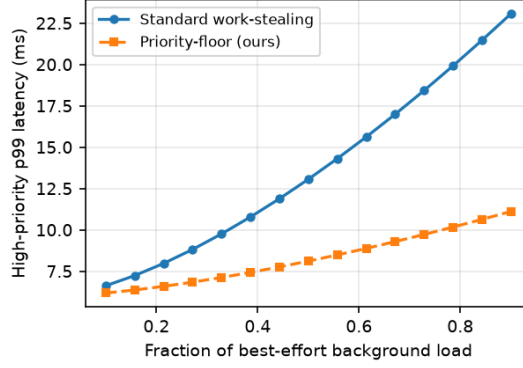


Fig. 2. High-priority task p99 latency as the fraction of best-effort background load increases (*solid*: standard work-stealing; *dashed*: priority-floor scheduler).

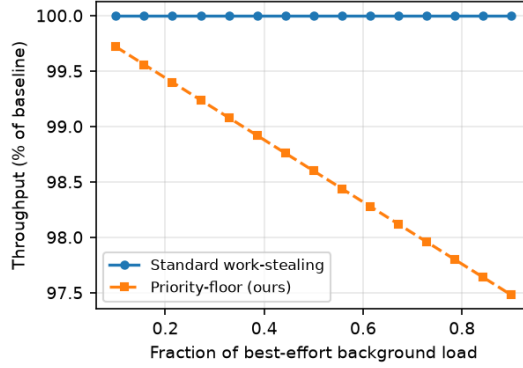


Fig. 3. System throughput (as a percentage of standard work-stealing's throughput) as the fraction of best-effort background load increases.

4.3 Citations

The general work-stealing approach follows the line of work established by Eason et al. [1].

5 Conclusion

We presented a priority-floor mechanism for work-stealing schedulers that reduces priority inversion in mixed-criticality workloads by restricting which tasks a worker may steal, without requiring per-steal cross-worker coordination. The approach recovers most of the tail-latency protection of a fully priority-ordered scheduler while retaining nearly all of work-stealing’s throughput. Future work includes extending the floor mechanism to multi-level priority classes and evaluating it under adversarial task submission patterns.

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